KOR STRUCTURAL — OPERATIONS

Backup Infrastructure — Target Design v2 & Build Plan

KOR-driven redesign · keeps T-Net licence + off-site · supersedes the 2026-07-23 v1 design

Date: 2026-07-23 **Owner:** Ian Lalonde (KOR) **Legend:** KOR T-NET TOUCHPOINT DONE

Goal. A streamlined, restore-proven backup environment KOR operates itself: fast file-level restore, instant VM recovery, healthy off-site replication, right-sized storage, and a deliberate **point-in-time archive** program. KOR leads the design and build; T-Net's Veeam licence and Cloud Connect off-site are retained as inputs, with T-Net kept informed rather than gating each step.

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1. Done to date COMPLETE

- Incident recovered; vCenter certificate re-accepted; all jobs healthy; email alerting live and verified.
- Veeam upgraded **12.0 → 12.3.2**. KOR-APP01 (Server 2025) added as mount server — **deduplicated file-level restore works.**
- KOR-FS01 protected: dedicated **forward-incremental** job → Synology. All three copies restore-tested.
- **~20 TB reclaimed on production** (retired BMZ-HO-FS007/DC005/APP002/RDS001; UC3200 7.5 → 27 TB free).
- Orphaned repository files cleared; ManualNew confirmed as a March-2026 point-in-time and retained as a fallback archive.

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2. Target architecture

Production (vSphere / UC3200) → **Veeam BK01 (12.3)**
① **Primary backup → Synology (F:)** — forward incremental + weekly synthetic full · mount server APP01
② **Off-site copy → Cloud Connect (T-Net Toronto)** — Backup Copy over the wire, KOR-built
③ **Archive tier → E: (repurposed)** — point-in-time fulls (ManualNew + 6-monthly exports) · immutable in phase 2

Design decisions:

- **Forward incremental + weekly synthetic full** across all jobs — measured ~4× faster and ~⅓ the space of reverse incremental here; resists chain corruption.
- **Synology = primary** (where FS01 + APP01 mount server already work). **E: = archive/second-copy tier** once its old FS01 chain retires (~23 TB back).
- **Three independent copies** of what matters: primary (Synology), off-site (Cloud Connect), point-in-time archive (E: → immutable). Classic 3-2-1.

3. Point-in-time archive program (new)

Why: normal retention only reaches back weeks. A slow ransomware intrusion or unnoticed corruption would poison every recent backup. A periodic point-in-time archive is the clean-room copy from *before* the problem — the best defence against the failure mode you can't see coming.

Mechanism

- **Export Backup (deliberate, on-demand):** right-click any restore point → *Export* → a standalone `.vbk` that is independent of any chain, never ages out, and restores on its own. This is exactly "pull one full out as a point-in-time." **Cadence: every 6 months** for FS01 (+ the small VMs — trivial size).
- **GFS retention (automatic alternative):** configure the job to auto-keep monthly/quarterly/yearly fulls. Set-and-forget; use if the manual ritual slips.

Rules that make archives safe

- **Off the active repository** — archives live on E: (the archive tier), never crowding the Synology's live backups.
- **Immutable (phase 2, the gold standard)** — an archive on object storage with object-lock, or a hardened Linux repository, cannot be deleted or encrypted for its retention period. Survives total ransomware compromise.
- **Retention:** suggest keeping 2 years of semi-annual archives (4 points). At ~5.8 TB per FS01 full that's ~23 TB — fits E: after the old chain retires, or moves to cheaper object storage.

ManualNew becomes archive #1 — the existing March-2026 full is your first point-in-time; it's renamed and relocated to E: in the build plan below.

4. Off-site — KOR-built, T-Net optional

The Cloud Connect connection already exists and works nightly (the `Kor-Replication` copy job). KOR can therefore build the FS01 off-site copy **without waiting on T-Net**: a new Backup Copy job, FS01 Synology chain → the existing Cloud Connect repo (11.5 TB free; FS01 is 5.8 TB — fits), seeded over the wire at the measured **182 Mbps**.

Approach	Initial seed (5.8 TB)	Ongoing
Full bandwidth, off-hours	**~3–4 days**	Daily incrementals ~20–40 min
Throttled ~50% for work hours	**~6–7 days**	

T-Net touchpoints (optional speed-ups, not gates): (a) does their Cloud Connect offer a target **WAN accelerator** to shrink the transfer; (b) do they support a shipped **seed-drive import** to skip the ~week WAN push. Either helps; neither is required — the wire works. Courtesy note to them that FS01 is being added to the off-site (uses more quota).

5. Build plan — ordered, KOR-driven

Phase A — Off-site copy of FS01 (the unlock) KOR OPTIONAL SPEED-UP

1. Home → **Backup Copy Job** → source: the `Kor-FS01` backup → target: `Kor-Replication` (Cloud Connect). Set a copy interval; enable compression.
2. **Throttle** for business hours (Backup Infrastructure → global network traffic rules, or the job's window) so it doesn't strangle the office link; full speed off-hours.
3. Start it. The initial ~5.8 TB seed runs over ~a week, resumable. Optionally ask T-Net re: WAN accelerator / seed-drive to shorten this.
4. Verify FS01 appears under Backups → Cloud with current restore points. **Off-site gap closed.**

Phase B — Reclaim E: (only after Phase A confirmed) KOR

1. Re-run restore tests: FS01 file-level from the Synology *and* from the new Cloud copy. Both must pass.
2. Home → Backups → Disk → the **old FS01 chain under** `Kor-VMs-New` → right-click → **Delete from disk**. Reclaims ~23 TB on E:.
3. E: is now the **archive / second-copy tier** with room.

Phase C — Archive program + relocate ManualNew KOR

1. **Relocate ManualNew to E:** — Home → Backups → Disk → right-click ManualNew → **Move backup** → target E: (frees ~12.7 TB on the Synology, preserves the archive). Rename the job to e.g. `Archive-2026-03`.
2. **Establish the 6-monthly ritual:** every 6 months, right-click FS01's latest restore point → **Export** → standalone `.vbk` to `E:\Archive\`. (Add the small VMs — trivial.) Label by date.
3. Optional automation: enable **GFS** on the FS01 job (keep quarterly/yearly fulls) instead of manual exports.

Phase D — Optimise & validate KOR

1. Convert `Kor-VMs-New` and `Vcenter` to **forward incremental + weekly synthetic full** (Edit job → Storage → Advanced → Backup).
2. **Validate Instant VM Recovery:** test-boot a small VM from backup via the APP01 mount server; confirm it runs, then discard. If FS01 IVMR matters for RTO, add a dedicated **cache disk to APP01** (its C: is only ~43 GB).
3. Finish the tidy-ups: retire APP003/APP004 on host .10 (they have backups — safe); resolve the `Kor-RDS01` / `Kor-RDS01_1` duplicate.

Phase E — Harden (phase 2, ongoing) KOR OPTION

1. **Immutability for archives** — object storage with object-lock (e.g., Wasabi/B2 capacity tier) or a hardened Linux repository. The ransomware-proof layer.
2. **Restore-test cadence:** quarterly file-level, annual full-VM / IVMR. Log each.
3. Optional: ReFS-64K on the backup volume for fast-clone (faster synthetic fulls, less space).

6. T-Net — what they're told, what they own

This document is shared with T-Net as the record of the redesign. KOR is driving; T-Net is kept informed, not asked to approve each step.

Informed of (courtesy, no sign-off needed):

- Veeam upgraded to 12.3.2; APP01 added as mount server; ~20 TB of retired VMs removed; the redesign to forward-incremental + archive tier; FS01 being added to the off-site copy (more Cloud quota used).

Genuine T-Net dependencies (retained by choice):

- **Veeam licence** — the Enterprise Plus rental. Confirm it covers continued use through these changes.
- **Cloud Connect off-site** — their Toronto endpoint. Optional asks: target WAN accelerator and/or seed-drive import to speed the FS01 seed; confirm quota headroom.
- **Monitoring/alerting SLA and vCenter certificate ownership** — agreed so an undetected failure can't recur (cert renews 2028-07-14).

KOR Structural Operations, 2026-07-23. Supersedes the v1 target design of the same date. Seed estimates based on a measured 182 Mbps office upload. Steps tagged KOR are executable without provider involvement; T-Net touchpoints are optional speed-ups or retained dependencies, not approvals. Shared with T-Net as the redesign record.